

LOSO Experiment B_ypr_lstm

LSTM only · YPR only (both hands), no Nothing

Generated 2026-05-11 19:15:15

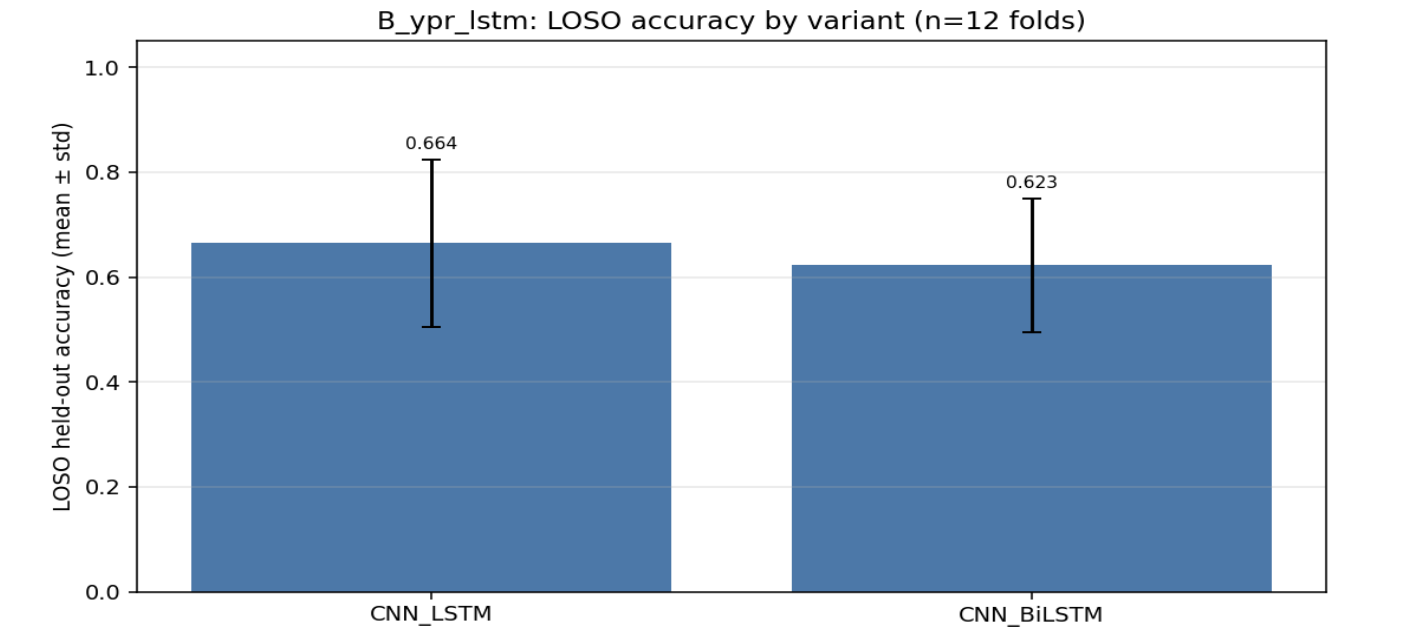
Experiment configuration

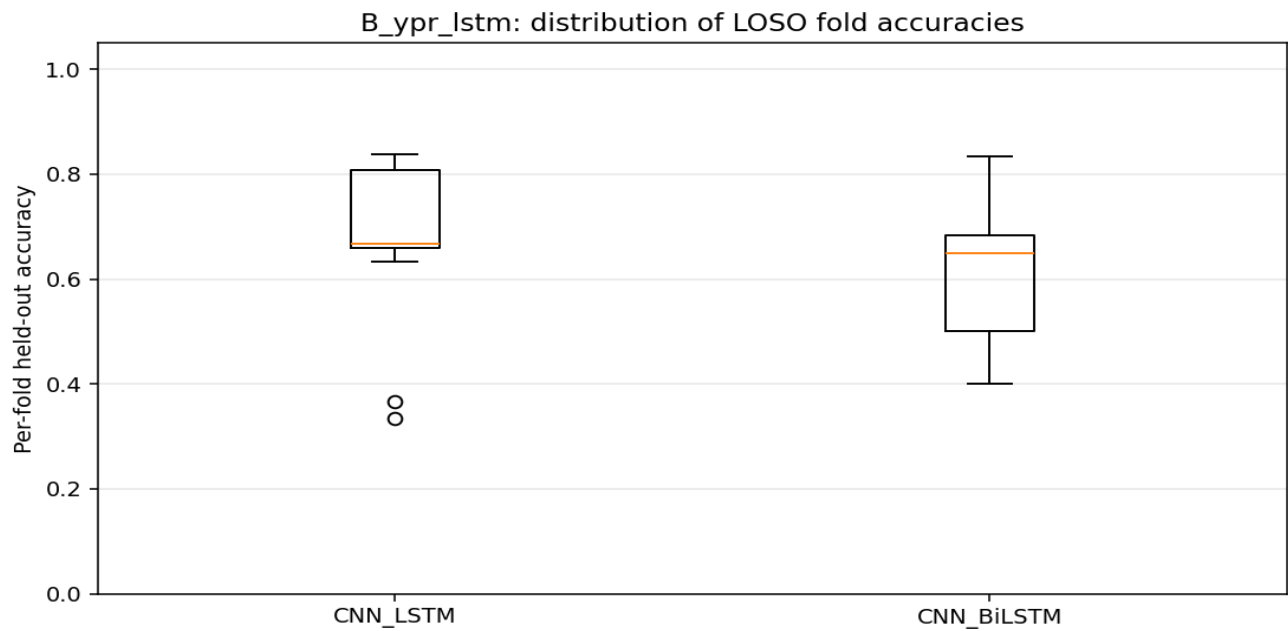
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	YPR
Hands	left, right
Sensor channels	78
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

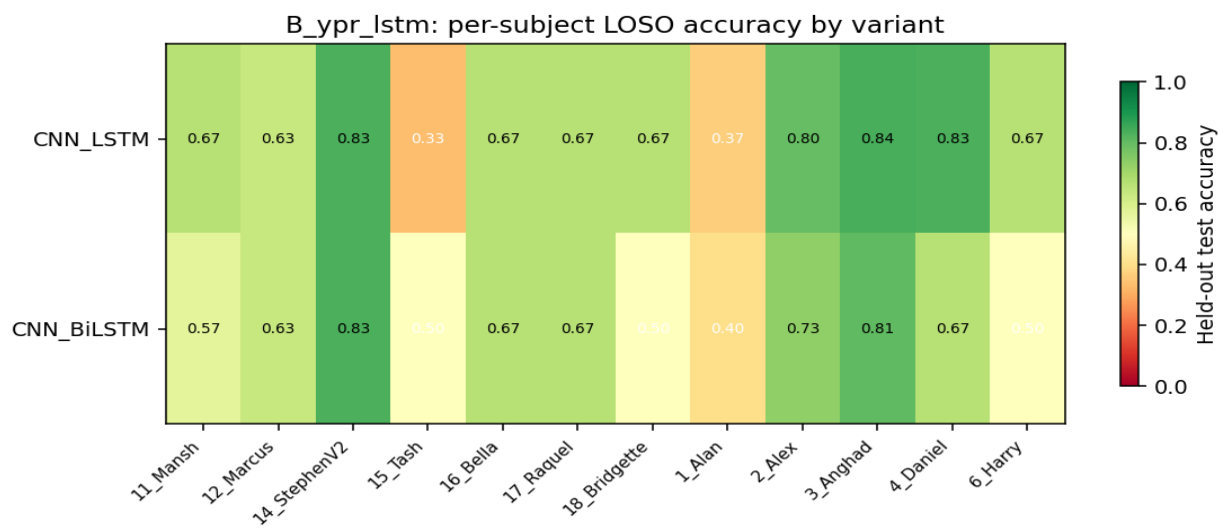
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	55,846	0.6643	0.1594	0.3333	0.8387	0.7593
CNN_BiLSTM	92,966	0.6228	0.1272	0.4000	0.8333	0.7501

Best variant by LOSO mean: **CNN_LSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.667	0.567
12_Marcus	331	0.633	0.633
14_StephenV2	331	0.833	0.833
15_Tash	331	0.333	0.500
16_Bella	331	0.667	0.667
17_Raquel	331	0.667	0.667
18_Bridgette	331	0.667	0.500
1_Alan	331	0.367	0.400
2_Alex	331	0.800	0.733
3_Anghad	330	0.839	0.806
4_Daniel	331	0.833	0.667
6_Harry	331	0.667	0.500

Variant: CNN_LSTM

Params: 55,846

LOSO mean accuracy: 0.6643 ± 0.1594 (min 0.3333, max 0.8387)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	0/5	5/5	0/5	0.667
12_Marcus	5/5	4/5	5/5	0/5	5/5	0/5	0.633
14_StephenV2	5/5	5/5	5/5	0/5	5/5	5/5	0.833
15_Tash	0/5	4/5	5/5	1/5	0/5	0/5	0.333
16_Bella	5/5	5/5	1/5	1/5	5/5	3/5	0.667
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
18_Bridgette	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	5/5	0/5	1/5	0/5	5/5	0/5	0.367
2_Alex	5/5	4/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	5/5	0/5	6/6	5/5	0.839
4_Daniel	5/5	5/5	5/5	0/5	5/5	5/5	0.833
6_Harry	5/5	5/5	5/5	0/5	0/5	5/5	0.667
MEAN	0.917	0.867	0.867	0.033	0.833	0.467	0.664

Variant: CNN_BiLSTM

Params: 92,966

LOSO mean accuracy: 0.6228 ± 0.1272 (min 0.4000, max 0.8333)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	2/5	5/5	0/5	5/5	0/5	0.567
12_Marcus	5/5	4/5	5/5	0/5	5/5	0/5	0.633
14_StephenV2	5/5	5/5	5/5	0/5	5/5	5/5	0.833
15_Tash	5/5	5/5	5/5	0/5	0/5	0/5	0.500
16_Bella	5/5	5/5	1/5	0/5	5/5	4/5	0.667
17_Raquel	0/5	5/5	5/5	5/5	5/5	0/5	0.667
18_Bridgette	5/5	5/5	0/5	0/5	5/5	0/5	0.500
1_Alan	5/5	2/5	0/5	0/5	5/5	0/5	0.400
2_Alex	5/5	3/5	5/5	0/5	5/5	4/5	0.733
3_Anghad	5/5	5/5	5/5	0/5	5/6	5/5	0.806
4_Daniel	5/5	5/5	4/5	0/5	5/5	1/5	0.667
6_Harry	5/5	4/5	1/5	0/5	4/5	1/5	0.500
MEAN	0.917	0.833	0.683	0.083	0.886	0.333	0.623